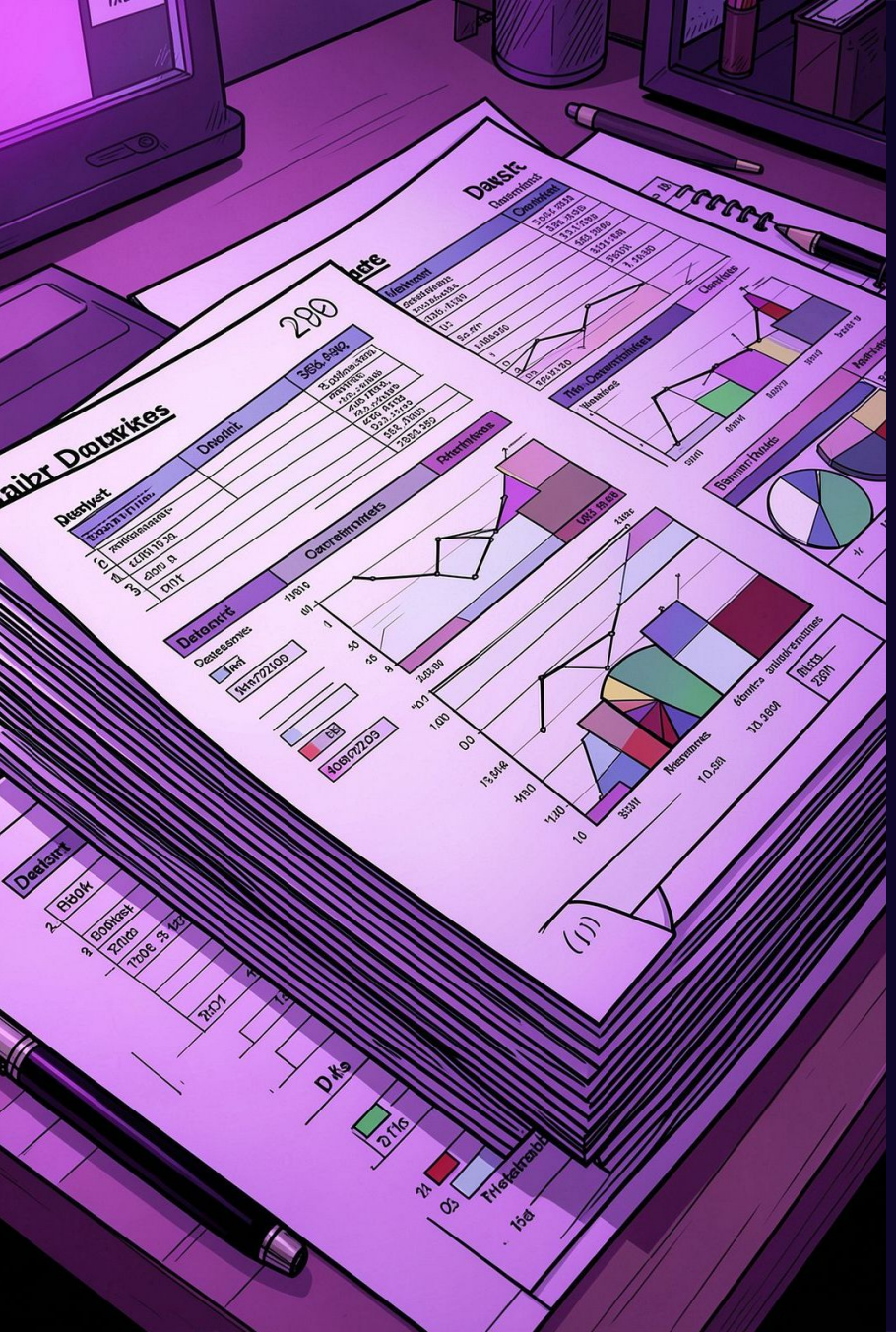


Customer Shopping Behavior — Executive Summary

Analysis of 3,900 purchases to reveal spending patterns, segments, product preferences, and subscription behavior. Insights guide subscriptions, promotions, and product positioning.

Made with GAMMA



Dataset Snapshot

Scale

3,900 rows × 18 columns

Key Features

Demographics, purchase details, behavior, shipping, discounts, ratings

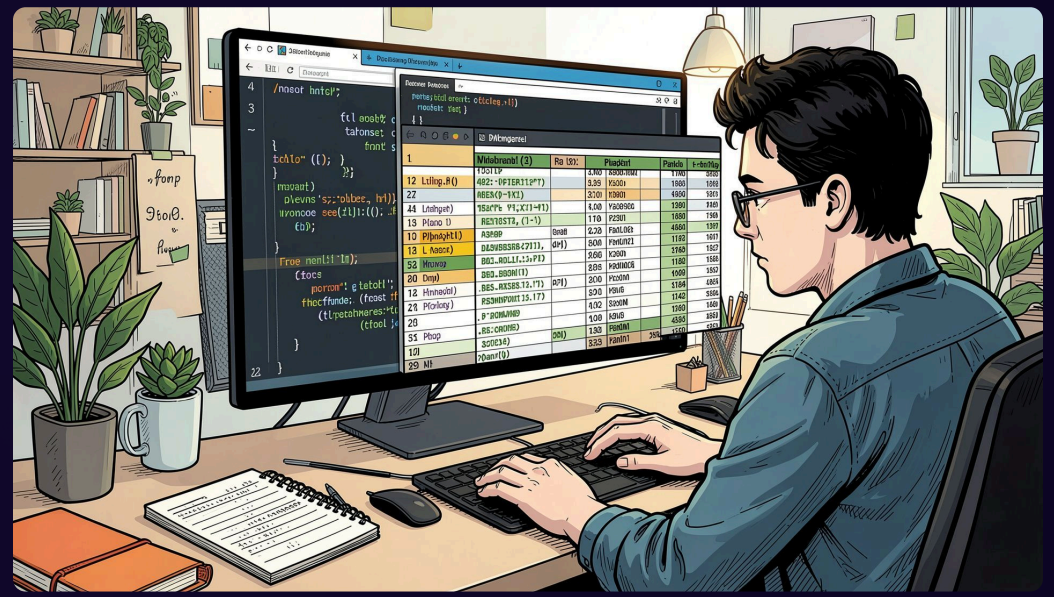
Missing Data

37 nulls in Review Rating (imputed by category median)

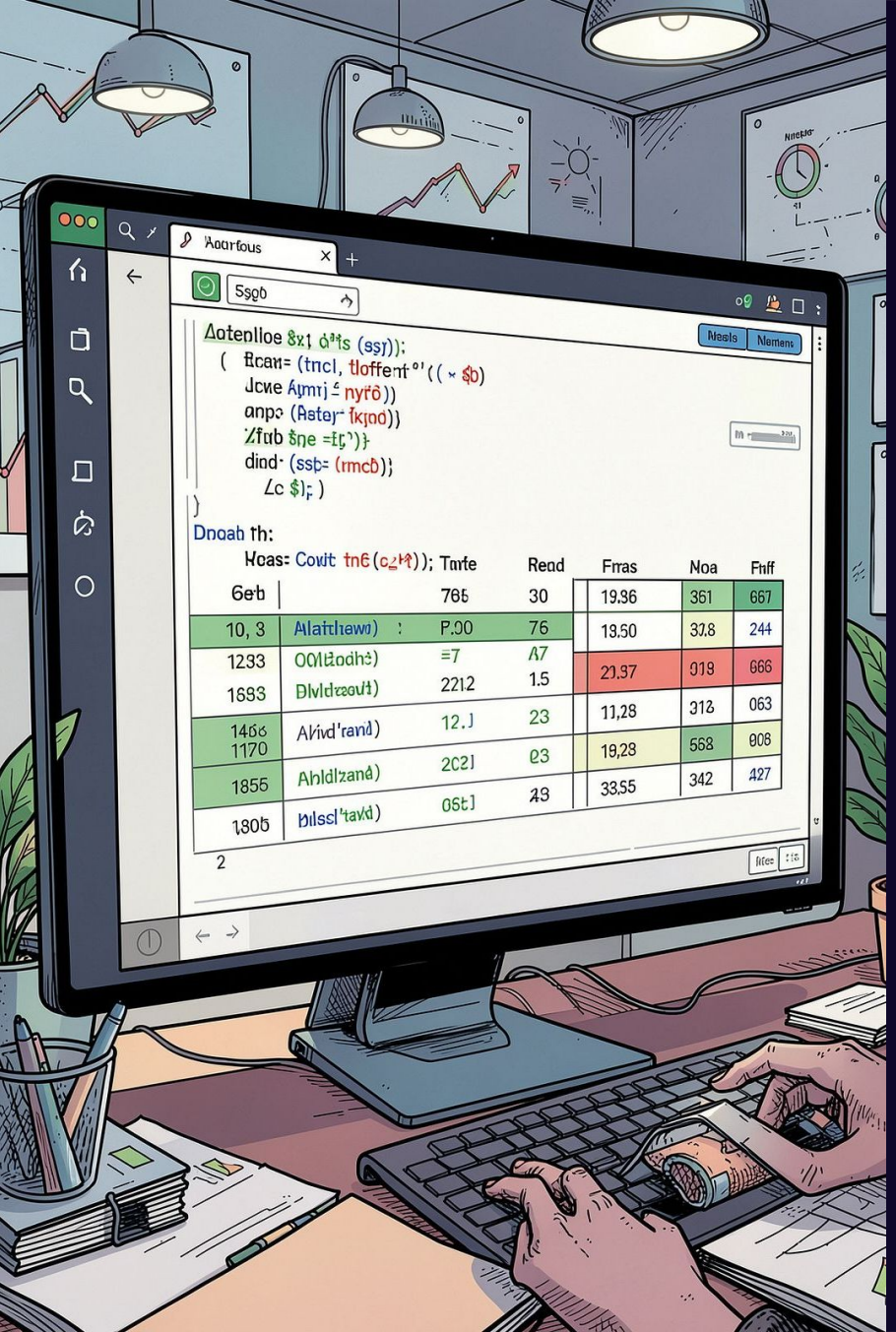
Made with GAMMA

Data Preparation (Python)

- Loaded with pandas; used df.info() and describe() for checks. - Imputed missing review ratings by product-category median. - Standardized column names to snake_case. - Engineered age_group and purchase_frequency_days. - Dropped redundant promo_code_used. - Loaded cleaned DataFrame to PostgreSQL.



Made with GAMMA



SQL Analysis — Key Business Queries

Revenue by Gender

Male: 157,890 — Female: 75,191

High-Spend Discount Users

Identified customers who used discounts yet spent above average (sample list extracted).

Top Rated Products

Top 5 by avg rating: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)

Made with GAMMA

Shipping & Subscription Insights



Shipping Comparison

Average purchase: Standard $58.46 — Express $60.48

Subscribers vs Non-Subscribers

Subscribers: 1,053 customers, avg $59.49, total $62,645. Non-subscribers: 2,847 customers, avg $59.87, total $170,436.

Made with GAMMA

Discount-Dependent Products

Products with highest share of discounted purchases — prioritize margin review and promo strategy:

1

Hat

Discount rate: 50.00%

2

Sneakers

49.66%

3

Coat

49.07%

4

Sweater

48.17%

5

Pants

47.37%

Made with GAMMA

Customer Segmentation & Loyalty



Segments

Loyal: 3,116 — Returning: 701 — New: 83

Repeat Buyers & Subscriptions

Repeat buyers (>5 purchases): 958. Subscription conversion among repeat buyers examined.

Made with GAMMA

Top Products & Category Leaders



Accessories

Top: Jewelry (171), Sunglasses (161), Belt (161)



Clothing

Top: Blouse (171), Pants (171), Shirt (169)



Footwear

Top: Sandals (160), Shoes (150), Sneakers (145)

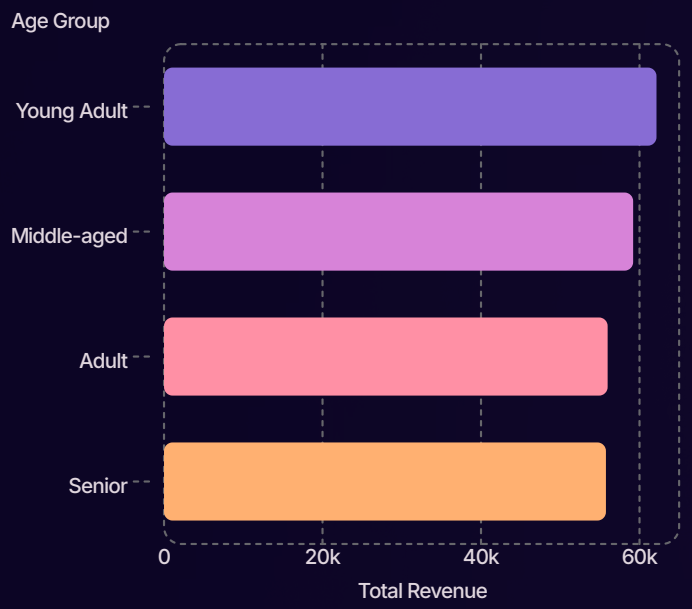


Outerwear

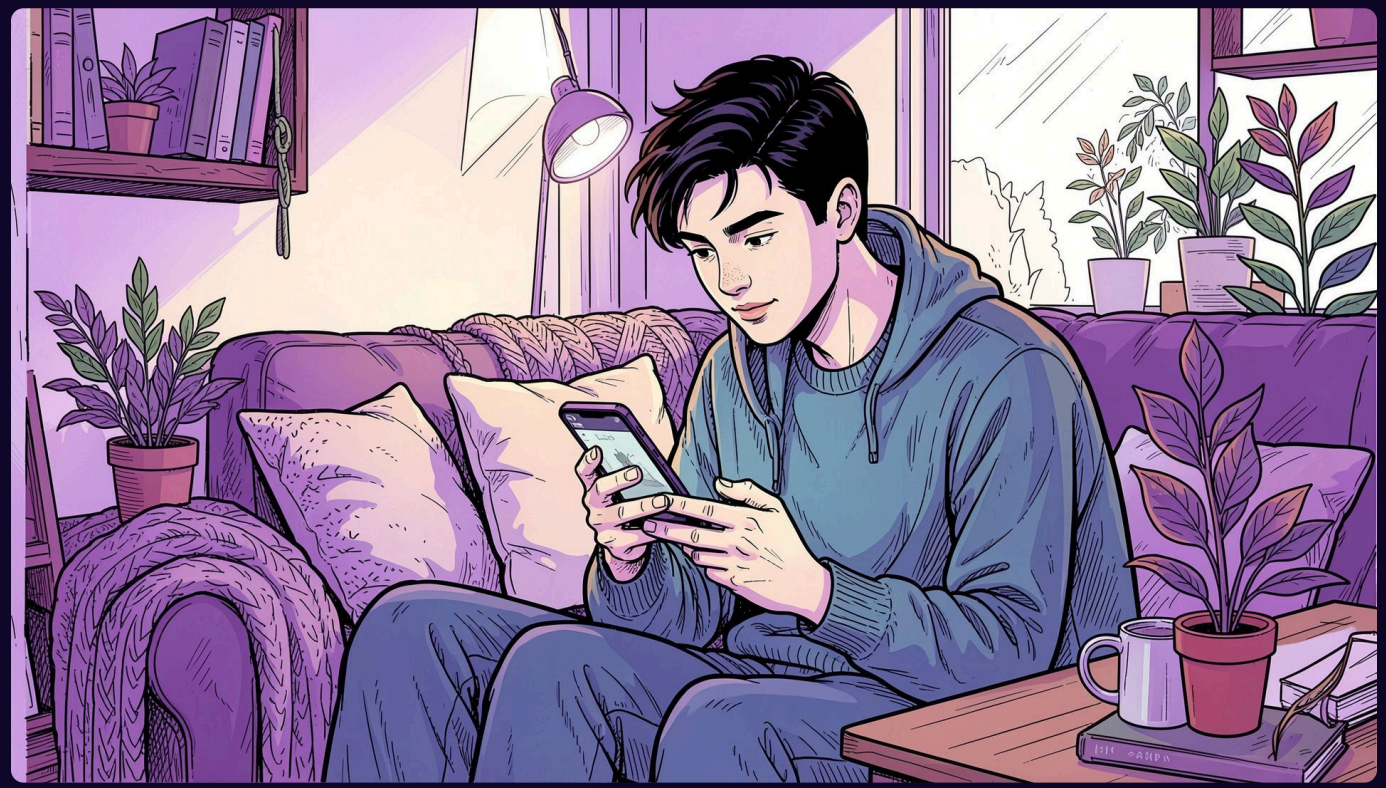
Top: Jacket (163), Coat (161)

Made with GAMMA

Revenue & Age Group Patterns



Young adults contribute the most revenue. Sales by age group also show Young Adult leading in order count — prioritize targeted campaigns for high-revenue cohorts.



Made with GAMMA

Recommendations & Next Steps

Boost Subscriptions

Market exclusive benefits to convert frequent buyers into subscribers.

Loyalty Programs

Reward repeat buyers to increase retention and move customers into the Loyal segment.

Review Discount Policy

Optimize promotions for high-discount items (Hat, Sneakers, Coat) to protect margins.

Product Positioning

Highlight top-rated and best-selling items in campaigns and product pages.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users for tailored offers.

Deliverables: dashboard updates, A/B tests for subscription offers, promo margin simulations, targeted campaign pilots.



Made with GAMMA